FAST-NUCES

National University of Computer and Emerging Sciences
Peshawar Campus

Computer Organization and Assembly Language Lab

LAB # 02

Direct and Indirect addressing

Student Name	Waleeja Khan	**Roll No.**	25P-0005
Section	BAI-3A	**Date**	11/Sep/2026
Instructor	Hamad Ahmad	**Marks**	____ / 10

Learning Objectives

- Understand immediate, direct, and indirect addressing in 8086.
- Access and modify data stored in memory.
- Use registers to hold memory addresses.
- Trace data movement between registers and memory.
- Predict register and memory values before execution.
- Verify program behavior using the debugger.

Instructions

1. Complete all tasks using DOSBox ,Nasm.
2. Use of AI-generated answers or solutions from classmates is strictly prohibited.
3. Use the debugger for instruction-by-instruction investigation where requested.
4. Answers and explanations should be written in the student's own words.
5. Save the completed document as: Lab1_<RollNo>_<Name>.docx

Task 1 :

If:

BX = 1000H

SI = 0020H

what memory address is accessed by:

```
MOV AX, [BX+SI]
```

Answer : 1020H

Task 2 :

Which is valid in 8086?

```
MOV AX, [BX]
MOV AX, [CX]
MOV AX, [SI]
MOV AX, [DI]
```

Explanation:

```
MOV AX, [BX]
MOV AX, [SI]
MOV AX, [DI]
```

These three instructions are valid because base and index registers can be used for addressing

Task 3 :

Explain the difference between:

```
MOV AX, BX
```

and:

```
MOV AX, [BX]
```

Answer:

MOV AX, BX copies the value present in BX to register AX.
While MOV AX, [BX] goes to the address present in BX and copies that value into AX.

Task 4 :

```
MOV BX, 2000H
MOV AX, BX
```

Memory:

Address	Value
2000H	55
2001H	66

What is the final value of AX?

Answer: 2000H

Task 5 :

```
MOV [1000H], [2000H]
After executing this what value is copied to Address 1000H?
```

Explanation: This instruction gives error because we can't use 2 addresses in one instruction. This is because of the manufacturing requirement of intel.

Task 6 :

```
MOV CX, 2500H
MOV AX, [CX]
```

1. What value is copied into AX?
   No value will be copied into AX, and this instruction will cause an error, because CX cannot be used for addressing.

Task 7 :

```
MOV BX, 3000H
MOV SI, 0002H
MOV AX, [BX+SI]
```

Memory Address	Value/Data
3000H	11
3001H	22
3002H	33
3003H	44

Don't run the program. Predict AX .

Write AX value: 33

Task 8 :

```
MOV BX, 1000H
MOV AL, [BX]
```

Memory:

1000H → 10

1001H → 20

1002H → 30

1003H → 40

Ask:

What is in AL?

Answer: 10

Then:

```
INC BX
MOV AL, [BX]
```

What is AL now? 20

Then:

```
INC BX
MOV AL, [BX]
```

Answer: 30

Task 09 :

How would you copy the contents of memory location 2000H to 1000H?

Answer:

[org 0x0100]

MOV SI, 2000H

MOV DI, 1000H

MOV word [SI], 5

MOV BX, [SI]

MOV [DI], BX

MOV AX, 4c00H

INT 21H

Task 10 :

A student's marks are **75**. Store the marks in memory location 2000H. Then retrieve the marks, add **5 bonus marks**, and store the final result in memory location 2001H.

Write and execute an 8086 Assembly Language program to perform this operation.

Before execution, predict the values of the registers and both memory locations. Then verify your predictions using the debugger.

Questions:

1. What will be the final value stored at 2000H?
2. What will be the final value stored at 2001H?
3. Which register is used to hold the memory address? AX
4. Trace how the value 75 moves through the registers and memory.

   Firstly, 75 was stored at memory address 2000h, then 5 was added into 75.

Prediction:

AX = 0050 BX = 2000 CX = 0000

[2000H] = 75 [2001H] = 80

Task Code:

```
[org 0x0100]

MOV BX, 2000H
MOV BYTE [BX], 75
MOV AL, [BX]
ADD AL, 5
MOV  [2001H], AL

MOV AX, 4c00H
INT 21H
```